

1 Results

Our central question was not whether a sufficiently flexible model could mimic human choices. It was whether the same choices were sufficient to identify how a learner searched, abandoned and eventually committed to a rule. Category learning makes this inverse problem especially acute: a correct response can reflect knowledge of the correct partition, a partially correct rule, or a fortuitous choice under an entirely different rule. We therefore designed the analysis to expose this ambiguity rather than conceal it. We first identified models that made comparably accurate choice predictions but implied different latent learning histories. We then asked whether oral reports— withheld from model fitting—could distinguish those histories. Only after this independent test did we use the inferred states to explain learning dynamics and ask whether the same process could generate them autonomously.

1.1 Similar learning outcomes concealed distinct paths through rule space

Ninety-six participants learned deterministic category structures built from the same four-dimensional stimuli. The three between-participant conditions separated elementary rule discovery from structured generalization: condition 1 required a binary division along one feature, whereas conditions 2 and 3 required a four-category hierarchical solution whose organization was either disclosed or left to be discovered. This design allowed endpoint performance to be separated from the route by which that endpoint was reached.

Participants did not approach the solution as uniformly noisy versions of one ideal learner. Aggregate accuracy improved in every condition [*Result pending: report condition-wise learning effects, uncertainty and between-participant dispersion*], but individual trajectories contained extended plateaux, abrupt gains, reversals and persistent misconceptions. [*Result pending: quantify change points, dwell times and the prevalence of abrupt versus gradual learning phenotypes.*] Importantly, similar accuracy curves could mask different paths: some participants repeatedly tested alternatives before a sudden transition, whereas others refined a stable but initially imprecise rule. Thus, the behavioral variation to be explained concerned the organization of learning, not merely its average rate.

Trialwise oral reports made that organization partly observable. Participants described structured relations—including feature magnitude, ordering, near-equality and central values—and often maintained a reported rule across several decisions before revising it. In the binary task, adding the recurrent near-equality and central-value relations to the initial rule catalogue yielded 29 candidate rules and increased coverage of structured reports from 84.5% to 92.8%. The oral data therefore supported a discrete but graded description of what participants considered: cognition moved among interpretable regions of rule space rather than along a single scalar measure of certainty.

The oral reports also contained information not recoverable from the current choice alone. Among trials matched on stimulus, choice, feedback history and current accuracy, the reported rule predicted subsequent rule revision and choice change [*Result*

pending: report conditional prediction or matched-trial effect with participant-level uncertainty]. This result is important for the logic that follows. The reports were neither treated as infallible access to mental state nor reduced to post-hoc explanations of a response; they provided a noisy second measurement of the evolving rule representation.

1.2 Behaviorally successful models disagreed about what learners believed

We next turned the inverse problem into a direct model comparison. The model set spanned two qualitatively different accounts. Continuous-accumulation models maintained and updated evidence across the available rule space. The dynamic rule model instead represented a capacity-limited active set, separated rules that were currently considered from the rule actually used for choice, and allowed feedback-dependent failure and mastery signals to control local refinement, global search and commitment. Nested variants removed these operations in turn, permitting the contribution of adaptive search, persistent execution and rule-specific confidence to be tested without changing the task representation.

All hyperparameters were selected from choices alone by coordinate-descent search (Hyper-CD) on the training portion of each participant’s sequence. The model structures and hyperparameters were frozen before forward prediction. Rather than declaring a winner from a small likelihood difference, we first defined a behavioral-equivalence set using held-out choice likelihood together with participant-level posterior-predictive checks of accuracy, volatility, switching and history dependence. [*Result pending: report the temporal split, Hyper-CD convergence, held-out model contrasts and the models retained in the behavioral-equivalence set.*]

Several models accounted for overt choices to a statistically and practically similar degree [*Result pending: report paired NLL differences, equivalence bounds and posterior-predictive coverage*], yet assigned sharply different probability to the rules a participant was considering on the same trial [*Result pending: report cross-model divergence of latent rule distributions*]. In particular, a model could reproduce when a participant was correct while misidentifying why. Choice prediction therefore constrained the possible learning processes but did not identify one of them.

This ambiguity was not resolved by selecting the single simulated trajectory that most resembled a participant. For every candidate model, particle-filtered (PF) inference approximated the distribution over all latent rule paths compatible with the preceding choices. Lines and uncertainty bands in the following analyses summarize this marginal distribution. The inferential object is consequently a trialwise distribution over possible internal states, not a retrospectively chosen narrative.

1.3 Withheld oral reports broke the behavioral equivalence

Behavioral equivalence created the decisive test of the model. Oral reports had not contributed to participant-level parameter estimation, PF weighting or model selection. We mapped each report to a likelihood over candidate rules and retained the latest report about each category, updating the reported category while preserving

the other category’s most recent value. Combining these category-specific likelihoods produced an oral-report distribution over the same rule space as the model. This representation avoided forcing an ambiguous statement into a single rule and allowed partial information to accumulate across trials.

We compared model and oral distributions at three levels. Full-distribution similarity tested whether probability mass was allocated to the same competing rules. Expected category centres tested agreement in the represented partition without requiring identical labels. Target-based analyses summed over all rules that implemented the reported category relation, rather than crediting only one arbitrarily designated target rule. Time-preserving permutations and held-out participants established how much alignment could arise from common progress through the task or from using an oral-informed rule vocabulary.

The dynamic rule model aligned more closely with the withheld oral distributions than did the behaviorally equivalent continuous and reduced models [*Result pending: report participant-level differences in full-distribution, expected-centre and target-equivalence alignment, with temporal-null and held-out-catalogue tests*]. Crucially, this advantage remained after matching models on held-out choice prediction [*Result pending: report the conditional or matched-model analysis*]. The evidence thus formed a double dissociation: overt behavior alone admitted multiple computational explanations, whereas the independent measurement of represented rules selected among them.

The currently completed fixed-configuration analysis establishes the scale of the measurable correspondence, although it is not a substitute for the Hyper-CD comparison. Across condition-1 participants, expected-centre similarity between the model and oral distributions averaged 0.885, and active-space Jensen–Shannon similarity averaged 0.478. Trialwise target-equivalence trajectories were positively associated [*Result pending: replace the current single-target diagnostic with the final equivalence-class estimate and its participant-level interval*]. These descriptive values show that the two channels converged, but the stronger mechanistic conclusion depends on the behavior-matched model contrast above.

The oral measure was treated as external evidence, not as ground truth. Its validity depended on three separable observations: reports were semantically related to the stimulus, stable when participants maintained a strategy and sensitive when behavior changed. [*Result pending: report these measurement-validity analyses and robustness to report omissions, parser uncertainty and category-specific carry-forward.*] This distinction makes the conclusion narrower but stronger: the selected model recovered the component of latent rule use that was shared by two noisy and independently analysed measurements.

1.4 Rule revision was punctuated and gated by the consequences of failure

Having established which latent distribution was psychologically constrained, we asked what it revealed about the learning algorithm. The dynamic model made a temporal prediction that a smooth evidence accumulator did not. Negative feedback should not cause an undifferentiated loss of confidence. Initially it should increase failure

pressure within the current neighbourhood, prompting local refinement; persistent failure should then reorganize the active set by recruiting distant rules. Only after this representational change should the executed rule, and therefore overt choice policy, switch. Conversely, sustained success should increase mastery, stabilize execution and suppress global search.

Event-locked analyses supported this sequence. Failure pressure rose before active-set replacement, active-set replacement preceded executed-rule switches, and those switches anticipated changes in oral reports and behavior [*Result pending: report event-locked time courses, lead-lag contrasts and participant-level uncertainty*]. Local substitutions dominated after isolated errors, whereas error streaks selectively increased global transitions [*Result pending: report transition-distance effects and their interaction with error-streak length*]. Following successful evidence, the probability of retaining the executed rule increased and rule uncertainty contracted [*Result pending: report mastery and dwell-time effects*].

These results distinguish an algorithmic account from a relabelling of the learning curve. The relevant model variables predicted independently measured transitions at their specified temporal positions, and reduced models lacking the corresponding control operation failed selectively at those events [*Result pending: report event-specific ablations*]. Human learning was therefore better characterized as punctuated, feedback-gated reorganization of a bounded rule workspace than as uniformly paced convergence on a fixed representation.

1.5 Task structure reorganized search rather than merely slowing learning

The hierarchy manipulation provided a test of whether the inferred mechanism generalized beyond the binary task. We kept the learning architecture fixed and changed only the rule library and information made available by the task. If the model merely absorbed idiosyncratic condition-1 choices, it should fail to predict how search was reorganized when the solution required a hierarchy.

Disclosing the hierarchy constrained global search and promoted staged local refinement within the relevant branch, whereas hiding the same hierarchy prolonged uncertainty over branches and increased returns to global search [*Result pending: report cross-condition effects on search distance, active-set entropy, branch errors and executed-rule dwell time*]. These differences explained condition effects on learning time beyond perceptual discriminability and mean response noise [*Result pending: report mediation or model-based decomposition, using the independent perceptual calibration*]. The same feedback-gated control law thus adapted its search to the structure of the problem rather than requiring a different learning algorithm for each task.

This comparison also tests a resource-rational interpretation of the dynamics. Participants facing greater independently measured uncertainty should require more evidence before commitment and should return to global search more often, but the direction and timing of these effects must follow from recoverable model quantities rather than from unconstrained parameter correlations. [*Result pending: report the preregistered relation to perceptual uncertainty and rule out learning-duration and task-condition confounds.*]

1.6 The validated process generated individual learning phenotypes autonomously

The preceding analyses conditioned inference on human choices. They could show that the model tracked latent states, but not that its learning process was sufficient to create a human-like learner. We therefore closed the explanatory loop with autonomous simulation. On each trial, the fitted model sampled its own category choice, received the feedback caused by that choice and updated its later state from its own history. No human choice was reintroduced after initialization.

The selected model reproduced not only mean accuracy but the joint distribution of time to criterion, abrupt and gradual improvement, misconception dwell, local volatility and rule-switch timing across participants [*Result pending: report autonomous posterior-predictive coverage and uncertainty for each preregistered phenotype*]. It also generated the observed coupling between oral-rule transitions and behavioral change when synthetic reports were sampled from the independently validated observation model [*Result pending: report this out-of-sample generative check*]. Reduced models made diagnostic errors: continuous accumulation converged too uniformly, removal of adaptive search eliminated prolonged misconceptions or recovery from them, and removal of persistent execution produced excessive trial-to-trial switching [*Result pending: report the model-by-phenotype failure matrix*].

Conditional trajectory replay is not this generative test. Replay continues to ingest a participant's choices and feedback and asks how latent paths would evolve under that observed history; autonomous simulation lets the model's own choices alter every subsequent state. The former is useful for diagnosing inference and appears only as a control, whereas the latter supports the claim that the proposed process can produce the behavioral phenomena it explains.

1.7 Recovery analyses bounded the level of mechanistic interpretation

Finally, we asked which parts of this account were identifiable from sequences of the present length. Structure recovery was evaluated before parameter recovery: data generated by each finalist model were passed through the entire Hyper-CD and model-selection pipeline, testing whether the architecture could be distinguished rather than whether one parameter could be recovered while the correct architecture was assumed. [*Result pending: report the cross-generating-model confusion matrix and the range of conditions in which the full structure is recoverable.*]

The targeted recovery analysis already completed for three control parameters showed why this distinction matters. Across 72 autonomously generated condition-1 trajectories and nine joint parameter profiles, the exact profile was recovered in 13 cases (18.1%; 95% Wilson interval, 10.9–28.5%), compared with a chance rate of 11.1%. Individual-parameter recovery ranged from 37.5% to 43.1%, with every interval including the one-in-three chance rate. We therefore do not interpret these fitted values as precise participant-level psychological traits. Parameter-level claims will be retained only where the final design demonstrates recovery; otherwise inference remains at the level of recoverable model structure, latent functions and event-level predictions.

PF likelihood calibration provides a second boundary. Candidate rankings were not stable under the completed 32–256-particle audit, so the formal Hyper-CD and likelihood-ranking results above remain placeholders until a stable particle budget or lower-variance estimator is established. This numerical control concerns confidence in model selection, not the cognitive distinction between a distribution over latent paths and a selected best trajectory.

Taken together, the intended evidence chain goes beyond showing that a model fits learning curves. Choices first reveal the underdetermination of the hidden learning process; a withheld oral channel then adjudicates among the competing processes; independently observed transitions test the proposed control law; task structure tests its generality; and autonomous simulation tests its generative sufficiency. On this account, individual category learning is not simply noisy convergence toward a known answer. It is the feedback-gated construction, testing and stabilization of a rule within a bounded cognitive workspace.

References